

Ultrasonic Avoiding

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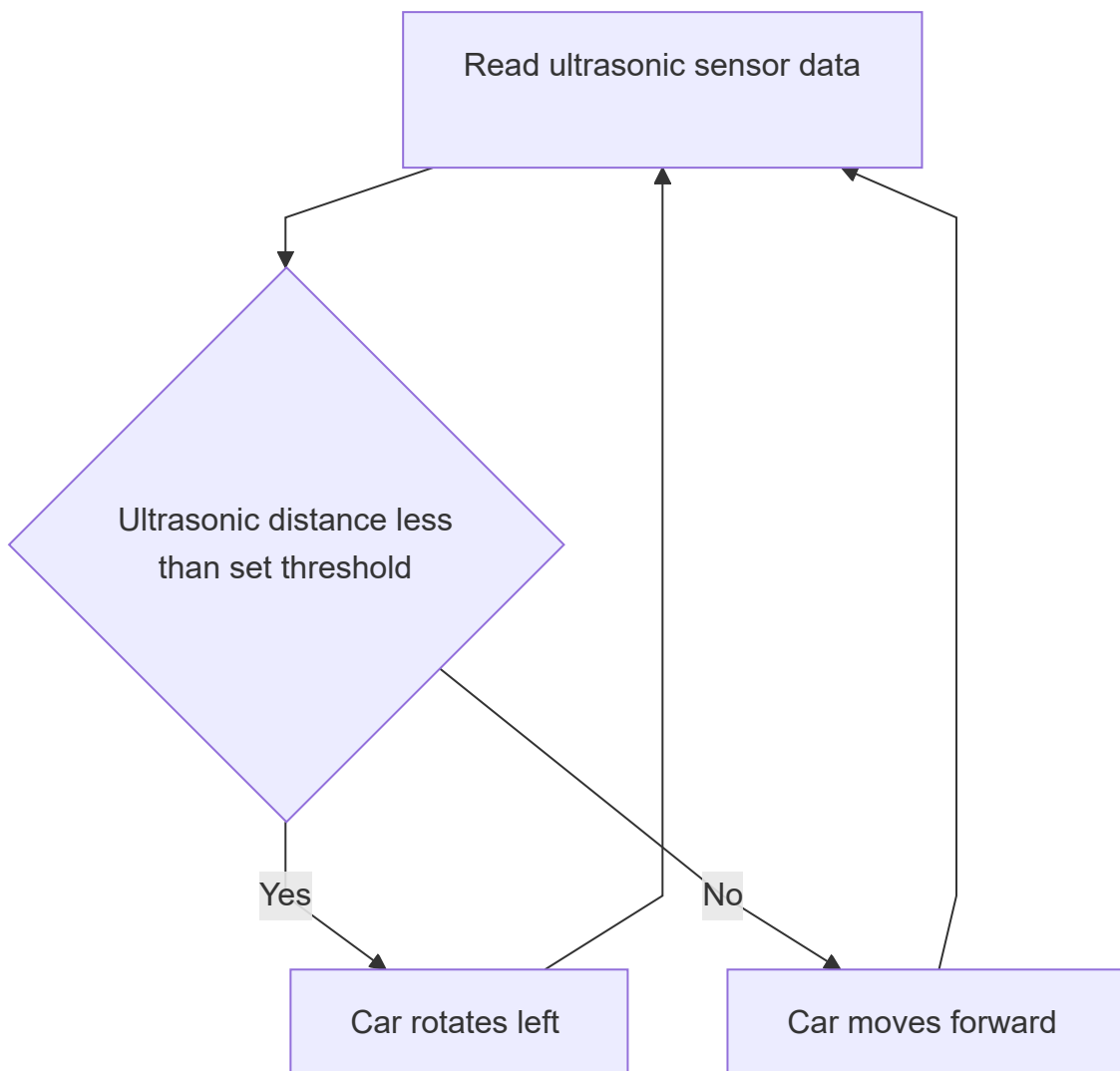
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Function Description

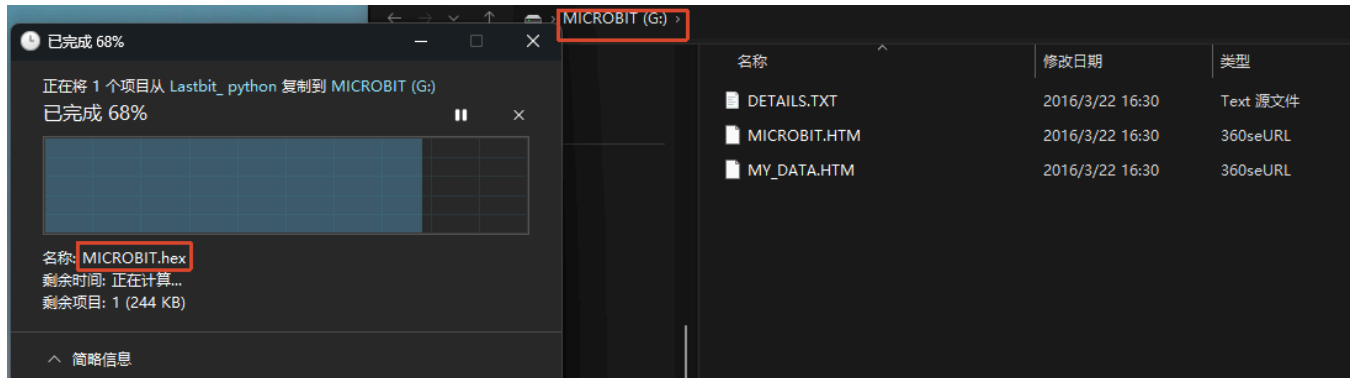
In this section, we will learn to use the ultrasonic sensor to detect obstacles and implement the automatic obstacle avoidance function. When an obstacle is detected ahead, the car will automatically turn left to avoid it.

Working Principle



Experiment Steps

Make sure `LASTBIT.hex` is flashed before flashing, otherwise an error will occur and the module cannot be imported



1. Before flashing, switch the switch to the OFF position and use a microUSB data cable to connect the computer and the micro:bit main board. **Note that it is the Microbit interface, not the Charging port on the expansion board.**
2. Open the Mu software. Here you can directly copy the program source code we provided; the operation steps are as follows. You can also enter the code manually in the editing window. Note! All English letters and symbols should be entered in English input mode. Use the Tab key for indentation, and the last line should end with a blank program.

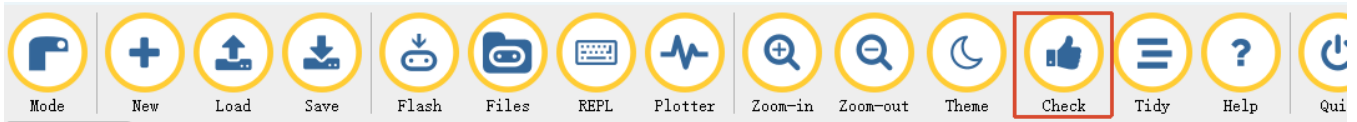
Click the `New` button in the toolbar at the top left, and a file titled "untitled" will appear.



3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a red dot after the title bar, indicating that the code content has been modified but is in an unsaved state.

Whether to save or not does not affect code checking or flashing.

4. Now you can click `Check` to verify whether there are any problems with the code format.



5. Now if you have already connected the MICROBIT to the PC in advance following the steps above, and flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently. When the editor shows `Copied code onto micro:bit.` at the bottom, it means the flashing is complete, and the indicator light will no longer blink after flashing.



6. Unplug the microUSB data cable and switch the switch to the ON position.

Code Implementation

```
# 16. Ultrasonic Obstacle Avoidance
from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

# Initialize servos
Lastbit.set_all_servos(90)
sleep(500)
Lastbit.set_servo('S2', 150)

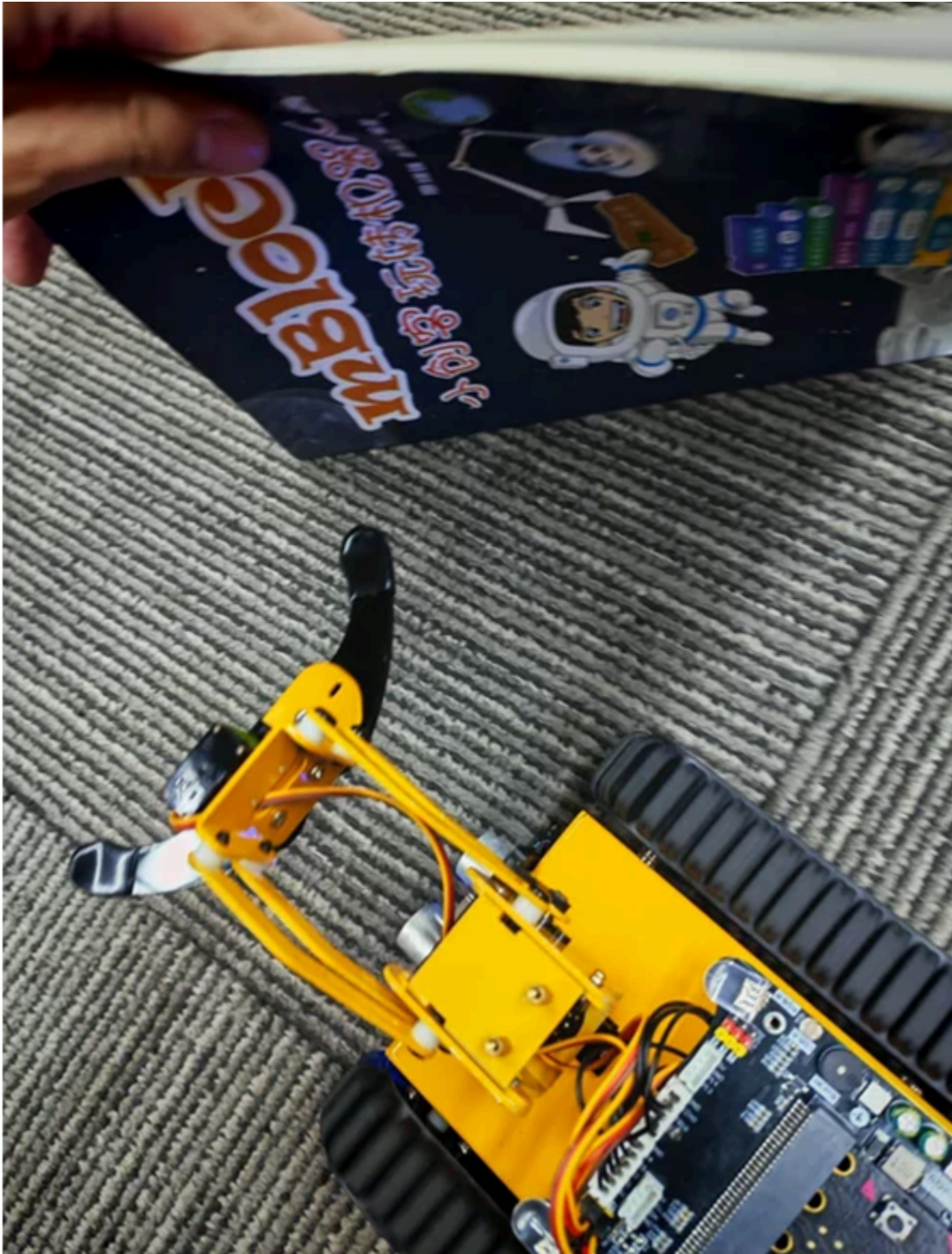
while True:
    # Read ultrasonic distance
    distance_mm = Lastbit.ultrasonic_distance()

    if distance_mm > 0 and distance_mm < 170:
        # Obstacle detected, turn left to avoid
        Lastbit.car_control('SPINLEFT', 170, 0)
    else:
        # No obstacle, move forward
        Lastbit.car_control('FORWARD', 70)

    sleep(50)
```

Experimental Phenomenon

After flashing the program, the car will move forward when there is no obstacle ahead, and rotate left when it encounters an obstacle.



Notes

- Make sure the ultrasonic sensor works properly. The distance detection threshold can be adjusted according to the actual situation.
- Make sure the battery is fully charged.